

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

### ASSESSMENT TOOL 1 – Concept Mapping

|                  |                                                                                                               |               |                                     |
|------------------|---------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | CSA0305                                                                                                       | Course Title: | Data Structures                     |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                    | Assessment:   | Assessment Tool 1 – Concept Mapping |
| Weightage:       | 35% of CIA                                                                                                    | Total Marks:  | 100                                 |
| CO1 Statement:   | Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. |               |                                     |
| Student Name:    | P. Harsha vardhan                                                                                             | Reg. No.:     | 192411179                           |
| Section / Batch: | CSE                                                                                                           | Date:         | 08/07/2026                          |

#### Code of Conduct

*p. Harsha Vardhan/ 192411179 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_\_\_\_

#### Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic                                                                   | Focus                                    | Q Nos | Marks     |
|-----------------------------------------------------------------------------------|------------------------------------------|-------|-----------|
| Linked data structure                                                             | Basic data structure                     | 1     | 20        |
| Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure | Classifications of Linear data structure | 2     | 20        |
| Search Data Structure                                                             | Linear Search and Binary Search          | 3     | 20        |
| Recursion, Performance analysis                                                   | Recursion                                | 4     | 20        |
| Tree Data Structures                                                              | Classifications of Tree data structure   | 5     | 20        |
| GRAND TOTAL:                                                                      |                                          |       | 100 Marks |

## Annexure A – Concept Mapping

1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates. (20 Marks)
2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing. (20 Marks)
3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)
4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them. (20 Marks)
5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data. (20 Marks)

### Rubrics for Calculation

| Criteria                         | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

# LINKED LIST VS ARRAY IN C

## ARRAYS



- ✓ Fast random access (indexing)
- ✓ Easy to use, memory continuous
- ✗ Fixed size
- ✗ Expensive insert/delete in middle

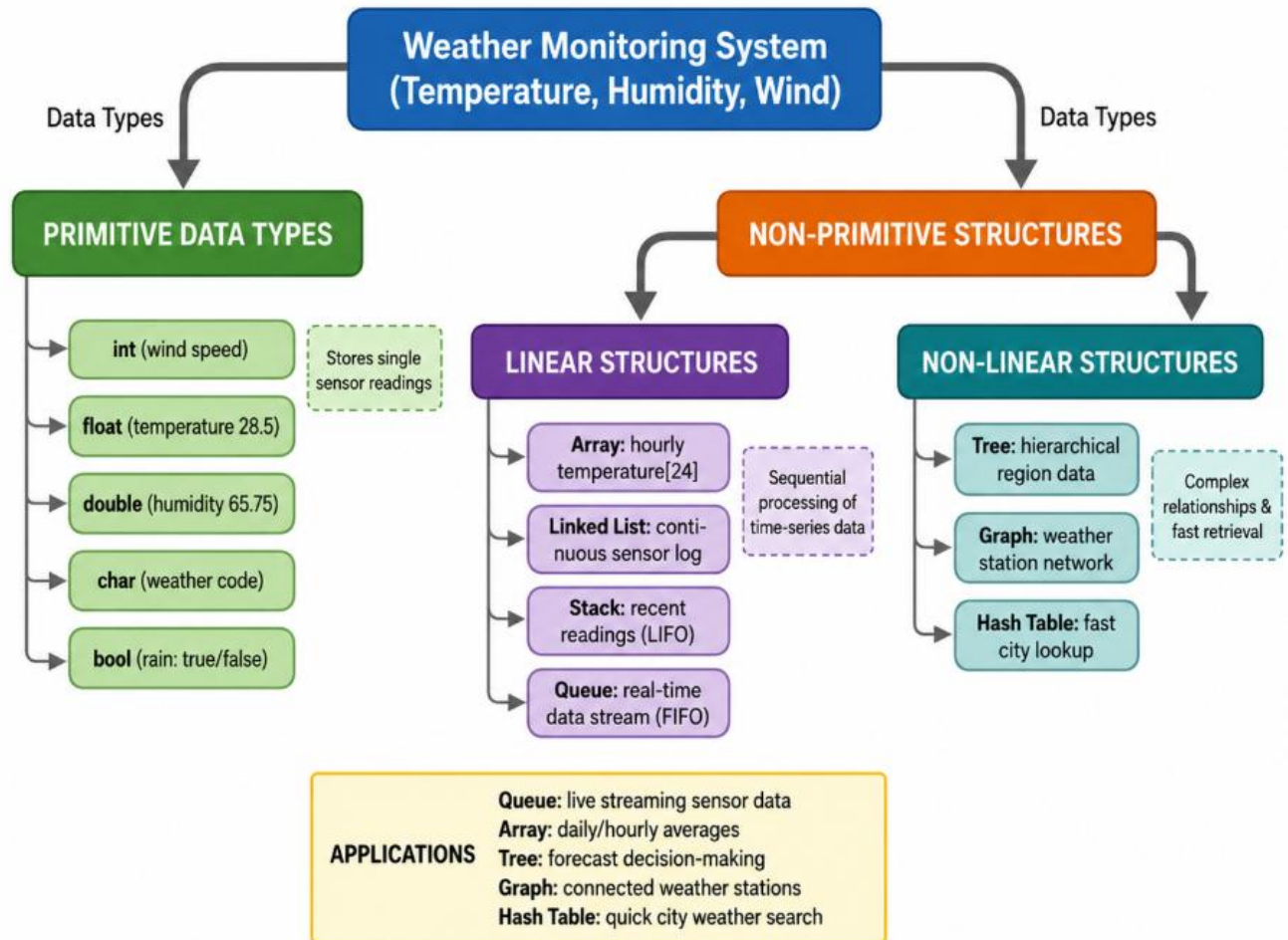
## LINKED LISTS



- ✓ Dynamic size (grow/shrink easily)
- ✓ Efficient insert/delete (just update pointers)
- ✗ Slower access (sequential search)
- ✗ Extra memory for pointers

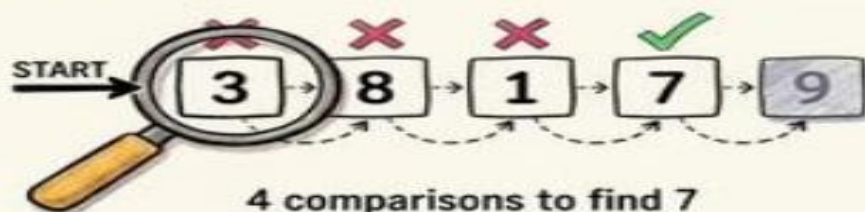
Linked Lists are useful when frequent insertions/deletions are needed.

## 2. WEATHER MONITORING SYSTEM - DATA STRUCTURES CONCEPT MAP



# LINEAR *VS* BINARY SEARCH

## LINEAR SEARCH



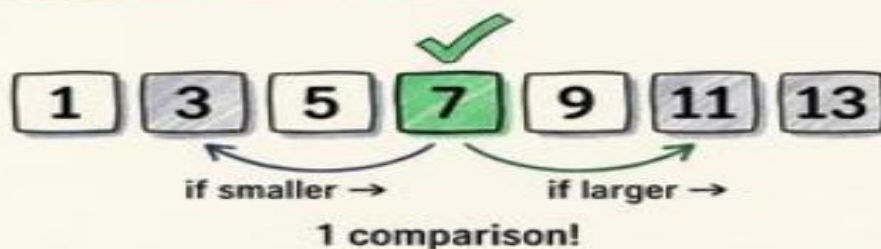
### HOW IT WORKS

- 1 Start at the first item
- 2 Compare each item to the target
- 3 Stop when found — or reach the end

Works on: ANY list  
(sorted or not)

Worst case: 1000 items  
→ 1000 comparisons

## BINARY SEARCH



### HOW IT WORKS

- 1 Find the middle item
- 2 Match? Done.
- 3 Target smaller? Search left half.
- 4 Target larger? Search right half.

Works on: SORTED  
lists only

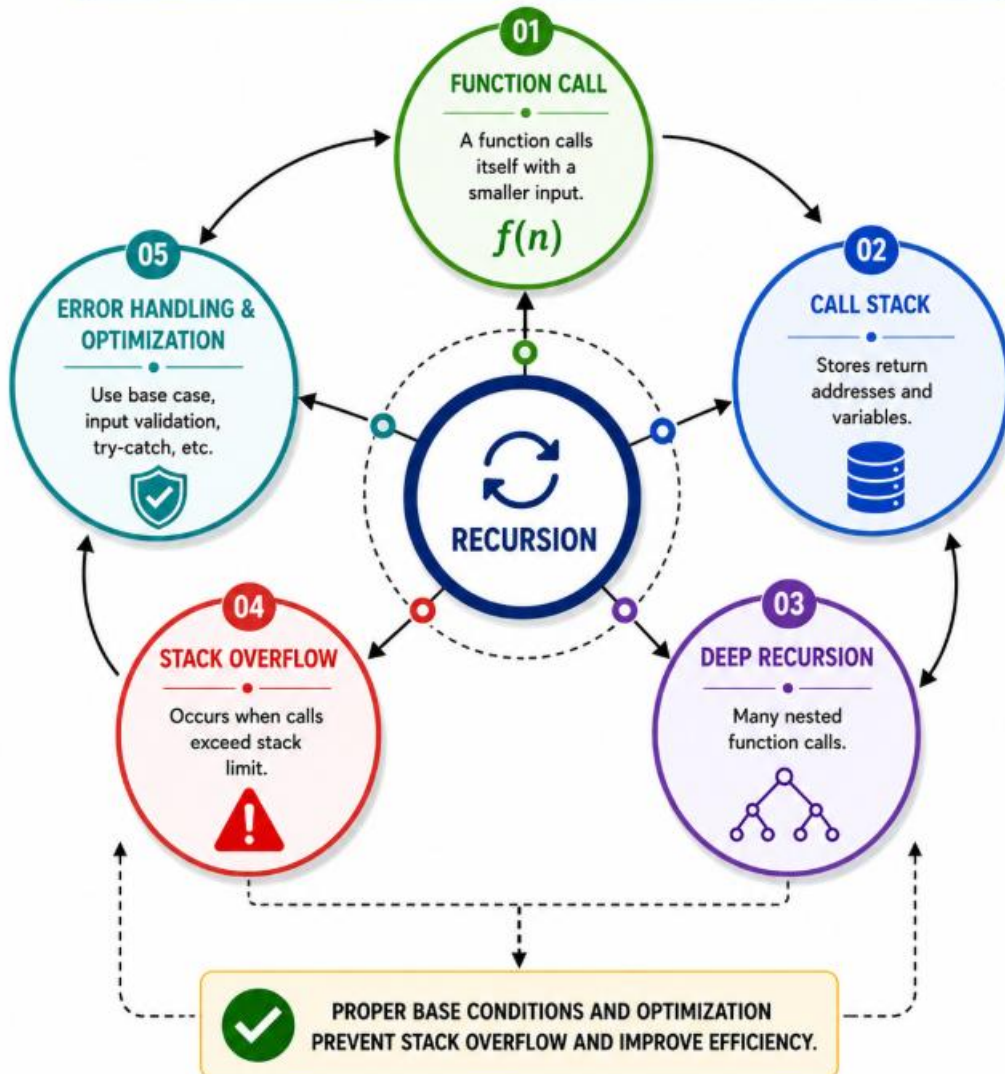
Worst case: 1000 items  
→ ~10 comparisons

|                        | LINEAR | BINARY |
|------------------------|--------|--------|
| Needs sorted list?     | NO     | YES    |
| 1000 items, worst case | 1000   | 10     |

The Adaptive Toolkit



## 4. RECURSION, PERFORMANCE ANALYSIS



## 5. TREE DATA STRUCTURES – DETAILS



- Each node represents an employee.
- It contains information like ID, Name, Designation, etc.

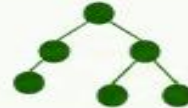


- Nodes are arranged in different levels.
- Level 0 is the top level (e.g., CEO).
- Lower levels represent subordinates.



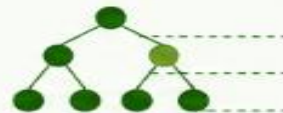
### **DFS** (Depth First Search)

- Goes deep into a branch first.
- Types: Preorder, Inorder, Postorder.



### **BFS** (Breadth First Search)

- Visits level by level.
- Uses a queue to explore all nodes at the current level first.



- Used to find a specific employee.
- Search by ID or Name.



- Performance depends on traversal choice.
- Affects time complexity and space used.



### **MOST SUITABLE APPROACH**

- ✓ **DFS** – Best for searching a specific branch or detailed information
- ✓ **BFS** – Best for level-wise search and shortest level information

